Complex Analysis

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November 8, 2026

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1 Complex Analysis (複分析 or 複變分析)

I Derivative

i Derivative

Let W be a topological vector space. The ordinary derivative or derivative of a function $f(u)\colon U \subseteq \mathbb{C} \to W$ (with respect to u) at $x \in U$, denoted as $f'(x)$, is defined as

$$f'(x) = \lim_{z \to x} \frac{f(z) - f(x)}{z - x}$$

if the limit exists. If such limit exists, we say f is (complex) differentiable at x.

We define the (first(-order)) ordinary derivative (function) or derivative (function) of f (with respect to u) as a function f' with codomain W such that for any $x \in U$ at which f is (complex) differentiable, f' maps x to the derivative of f at x.

The derivative of the kth(-order) ($k \in \mathbb{N}$) derivative function of f at $x \in U$ is called the $(k+1)$th(-order) derivative of f (with respect to u) at $x \in U$. The derivative function of the kth(-order) ($k \in \mathbb{N}$) derivative function of f is called the $k+1$th(-order) derivative function of f (with respect to u).

If for any $n \in \mathbb{N}$, the nth(-order) derivative of a function f at a point x in its domain exists, we say f is infinitely (complex) differentiable at x.

If f is (complex) differentiable at all point in $I \subseteq U$, we say f is (complex) differentiable on I; if f is (complex) differentiable on U, we say f is (complex) differentiable. If $f^{(n-1)}$ exists and is (complex) differentiable at all point in $I \subseteq U$, we say f is n-times (complex) differentiable on I; if $f^{(n-1)}$ exists and is (complex) differentiable on U, we say f is n-times (complex) differentiable.

The operation of finding the derivative or derivative function is called ordinary differentiation or differentiation.

Specifically, the 0th(-order) derivative of f is f itself.

ii Smoothness

Same as that of real.

iii Cauchy–Riemann equations

A function $f(x+iy) = u(x,y) + iv(x,y)$ for some real-valued functions u, v with $D_u, D_v \subseteq \mathbb{R}_2$ is holomorphic if u and v satisfy the Cauchy–Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y},$$

$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

The converse is true if f is continuous.

Proof. TODO □

iv Holomorphic or analytic function

A function f is holomorphic or analytic on $U \subseteq D_f$ if it is complex differentiable at every $z \in U$.

A function f is holomorphic or analytic if it is holomorphic on D_f.

A function f is holomorphic or analytic at $z \in D_f$ if it is holomorphic on some open neighborhood of z.

v Biholomorphic function

A function f is biholomorphic on $U \subseteq D_f$ iff f_U is bijective and holomorphic and $(f_U)^{-1}$ is holomorphic.

A function is biholomorphic iff it is bijective and holomorphic and its inverse is holomorphic.

II Poles and Zeros

i Poles

An isolated singularity z_0 of a complex function $f(z)$ is a pole of it iff there exists $n \in \mathbb{N}$, called the order, multiplicity, or degree of z_0, such that

$$(z - z_0)^n f(z)$$

is holomorphic on an open neighborhood of z_0 and

$$\lim_{z \to z_0} \left(z - z_0\right)^n f(z) \neq 0.$$

A simple or single pole is a pole of order 1. A repeated or multiple pole is a pole of order > 1.

ii Zeros

A point is a zero of a complex function f iff it is a pole of $\frac{1}{f}$.

A point is a zero of order, multiplicity, or degree $-n$ of a complex function f iff it is a pole of order n of $\frac{1}{f}$.

A simple or single zero is a zero of order -1. A repeated or multiple zero is a zero of order < -1.

iii Meromorphic function

A complex function f is meromorphic on an open set U if for all $z \in U$, either f is holomorphic at z or z is a pole of f.

A complex function f is meromorphic if it is meromorphic on D_f.

iv Pole-zero constellation or plot

A scattering plot on the complex plane showing the poles and zeros of a meromorphic function.

III Essential singularity

An isolated singularity of a complex function is an essential singularity iff it is neither a removable singularity nor a pole.

IV Annulus

An annulus $\operatorname{ann}(a; r, R)$ with $R \leq \infty$ in the complex plane is a region defined as

$$r < |z - a| < R.$$

If $r = 0$, the region is also called punctured disk.

V Contour

A contour is a parameterized curve $\gamma(t) : [a, b] \to \mathbb{C}$ that is piecewise C^1.

VI Contour integral

The contour integral of a complex function f defined on the image of a contour $\gamma(t) : [a, b] \to \mathbb{C}$ over γ is defined as the Riemann–Stieltjes integral:

$$\int_\gamma f(z)\,\mathrm{d}z := \int_{t=a}^{b} f(\gamma(t))\,\mathrm{d}\gamma(t)\,.$$

i Cauchy integral theorem or Cauchy–Goursat theorem

TODO

ii Cauchy's integral formula

TODO

iii Laurent series

Let f be holomorphic on an annulus $A = \operatorname{ann}(z_0; r; R)$. The Laurent series of f at z_0 is

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}, \quad z \in A,$$

where the first term is called regular or analytic part and the second term is called principal or singular part, and a_n is given by

$$a_n = \frac{1}{2\pi i} \oint_\gamma \frac{f(z)}{(z - z_0)^{n+1}} \, \mathrm{d}z$$

for any closed contour $\gamma \in A$ enclosing z_0.

- $\forall n \in \mathbb{N} : \ a_{-n} = 0$: either f is holomorphic at z_0 or z_0 is a removable singularity of f.
- $\exists N \in \mathbb{N}$ s.t. $a_{-N} \neq 0 \wedge \forall n \in \mathbb{N}_{>N} : \ a_{-n} = 0$: z_0 is a pole of order N of f.
- Otherwise: z_0 is a essential singularity of f.

a_{-1} is called residue of f at z_0, denoted as $\operatorname{Res}(f, z_0)$ or $\operatorname*{Res}_{z_0}(f)$, that is,

$$\operatorname{Res}(f, z_0) = \frac{1}{2\pi i} \oint_\gamma f(z) \, \mathrm{d}z \, .$$

iv (Cauchy's) residue theorem

Let U be a simply connected open subset of the complex plane containing a finite list of points $a_1, \ldots, a_n$, $U_0 = U \setminus \{a_1, \ldots, a_n\}$, and a function f is holomorphic on U_0. Let γ be a closed rectifiable curve in U_0. Let the winding number of γ around a_k be $I(\gamma, a_k)$.

$$\oint_\gamma f(z) \, \mathrm{d}z = 2\pi i \sum_{k=1}^{n} I(\gamma, a_k) \operatorname{Res}(f, a_k).$$

Proof. By divergence theorem, obviously. □

v Jordan's lemma

Consider a semicircular contour

$$C_R = \{ R e^{i\theta} : \ \theta \in [0, \pi] \}$$

If $f(z)$ is a function of the form

$$f(z) = e^{iaz} g(z), \quad z \in \mathbb{C}$$

with a positive parameter $a > 0$, then Jordan's lemma states that

$$\left| \int_{C_R} f(z) \, \mathrm{d}z \right| \leq \frac{\pi}{a} M_R,$$

where

$$M_R \coloneqq \max_{\theta \in [0,\pi]} \left| g(R e^{i\theta}) \right| .$$

Proof.

$$\begin{aligned}
\left|\int_{C_R} f(z)\,\mathrm{d}z\right| &= \left|\int_0^{\pi} g(Re^{i\theta})e^{iaR(\cos\theta+i\sin\theta)}iRe^{i\theta}\,\mathrm{d}\theta\right| \\
&= R\left|\int_0^{\pi} g(Re^{i\theta})e^{aR(i\cos\theta-\sin\theta)}ie^{i\theta}\,\mathrm{d}\theta\right| \\
&\le R\int_0^{\pi}\left|g(Re^{i\theta})e^{aR(i\cos\theta-\sin\theta)}ie^{i\theta}\right|\mathrm{d}\theta \\
&= R\int_0^{\pi}\left|g(Re^{i\theta})\right|e^{-aR\sin\theta}\,\mathrm{d}\theta \\
&\le RM_R\int_0^{\pi} e^{-aR\sin\theta}\,\mathrm{d}\theta \\
&= 2RM_R\int_0^{\frac{\pi}{2}} e^{-aR\sin\theta}\,\mathrm{d}\theta\,, \quad \sin\theta\ge\frac{2\theta}{\pi} \\
&\le 2RM_R\int_0^{\frac{\pi}{2}} e^{-\frac{2aR\theta}{\pi}}\,\mathrm{d}\theta \\
&= 2RM_R\left\{-\frac{\pi}{2aR}e^{-\frac{2aR\theta}{\pi}}\right\}\Bigg|_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{a}M_R\left(1-e^{-aR}\right) \\
&\le \frac{\pi}{a}M_R
\end{aligned}$$

□

If f is integrable on C_R for all sufficiently large R and

$$\lim_{R\to\infty} M_R = 0$$

then by Jordan's lemma

$$\lim_{R\to\infty}\int_{C_R} f(z)\,\mathrm{d}z = 0$$

vi Estimation lemma

TODO